In-Class Practice: Nonlinear PK

Module 8 - Nonlinear Pharmacokinetics · In-class activity · 14 Oct 2026

Name _____ Date _____

Question 1 What daily dose rate of a non-linear drug would provide a steady-state concentration of 15 mg/L? In this patient, the KM is 6.9 mg/L and the Vmax is 584 mg/day.

Question 2 When a 25 year-old, 75 kg patient was taking a daily oral dose of 350 mg, his steady state plasma concentration was 9.2 mg/L. When the daily dose was increased to 450 mg, a steady-state plasma concentration of 20 mg/L resulted. What is the KM of this drug in this patient?

Question 3 2. When a 25 year-old, 75 kg patient was taking a daily oral dose of 350 mg, his steady state plasma concentration was 9.2 mg/L. When the daily dose was increased to 450 mg, a steady-state plasma concentration of 20 mg/L resulted. 3. Based on the KM found in the previous question, what is the patient's Vmax?

4a 4. A 66-kg patient is receiving phenytoin for the treatment of seizures. When she was taking a daily oral dose of 300 mg phenytoin, her steady-state plasma concentration was 7.2 mg/L. Because the phenytoin plasma concentration was well below the therapeutic range, the patient's dose was increased to 500 mg phenytoin, which resulted in a steady-state plasma concentration of 18 mg/L. (Assume that the absolute bioavailability of phenytoin is 100% and the volume of distribution is 0.7 L/kg.) a. Calculate the patient's KM

4b 4. A 66-kg patient is receiving phenytoin for the treatment of seizures. When she was taking a daily oral dose of 300 mg phenytoin, her steady-state plasma concentration was 7.2 mg/L. Because the phenytoin plasma concentration was well below the therapeutic range, the patient's dose was increased to 500 mg phenytoin, which resulted in a steady-state plasma concentration of 18 mg/L. (Assume that the absolute bioavailability of phenytoin is 100% and the volume of distribution is 0.7 L/kg.) b. Calculate the patient's Vmax

4c 4. A 66-kg patient is receiving phenytoin for the treatment of seizures. When she was taking a daily oral dose of 300 mg phenytoin, her steady-state plasma concentration was 7.2 mg/L. Because the phenytoin plasma concentration was well below the therapeutic range, the patient's dose was increased to 500 mg phenytoin, which resulted in a steady-state plasma concentration of 18 mg/L. (Assume that the absolute bioavailability of phenytoin is 100% and the volume of distribution is 0.7 L/kg.) c. Recommend a daily dose of phenytoin for this patient to achieve a phenytoin plasma concentration of 15 mg/L at steady state.

4d 4. A 66-kg patient is receiving phenytoin for the treatment of seizures. When she was taking a daily oral dose of 300 mg phenytoin, her steady-state plasma concentration was 7.2 mg/L. Because the phenytoin plasma concentration was well below the therapeutic range, the patient's dose was increased to 500 mg phenytoin, which resulted in a steady-state plasma concentration of 18 mg/L. (Assume that the absolute bioavailability of phenytoin is 100% and the volume of distribution is 0.7 L/kg.) d. Calculate the half-life and the total body clearance when the plasma concentration is 15 mg/L. What is the half-life?

4d 4. A 66-kg patient is receiving phenytoin for the treatment of seizures. When she was taking a daily oral dose of 300 mg phenytoin, her steady-state plasma concentration was 7.2 mg/L. Because the phenytoin plasma concentration was well below the therapeutic range, the patient's dose was increased to 500 mg phenytoin, which resulted in a steady-state plasma concentration of 18 mg/L. (Assume that the absolute bioavailability of phenytoin is 100% and the volume of distribution is 0.7 L/kg.) d. Calculate the half-life and the total body clearance when the plasma concentration is 15 mg/L. What is the total body clearance?

4e 4. A 66-kg patient is receiving phenytoin for the treatment of seizures. When she was taking a daily oral dose of 300 mg phenytoin, her steady-state plasma concentration was 7.2 mg/L. Because the phenytoin plasma concentration was well below the therapeutic range, the patient's dose was increased to 500 mg phenytoin, which resulted in a steady-state plasma concentration of 18 mg/L. (Assume that the absolute bioavailability of phenytoin is 100% and the volume of distribution is 0.7 L/kg.) e. What steady-state concentration should be achieved if the dose were 400 mg daily?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

Question 1 Worked by the drill

Answer: 400 mg/day

WHAT THE STEM GIVES

C_{ss}	15 mg/L	the target, top and bottom
KM	6.9 mg/L	added to C_{ss} in the denominator
Vmax	584 mg/day	the numerator; units of R

THE WORKING, ONE LINE AT A TIME

SETUP

$$R = \frac{Vmax \times C_{ss}}{KM + C_{ss}}$$

KM and Vmax are given, and the target C_{ss} is 15 mg/L, so the nonlinear dose-rate line answers directly. Vmax is already in mg/day, so no weight is needed.

ALGEBRA

$$R = \frac{(584\ mg/day)(15\ mg/L)}{(6.9 + 15)\ mg/L} = \frac{8760}{21.9} = 400.0\ mg/day$$

mg/L cancels top and bottom and leaves mg/day, the units of Vmax, so R comes out as a daily dose rate. The target C_{ss} appears on top and again, added to KM, on the bottom, which keeps R below Vmax for any target.

ROUND

$$R = 400\ mg/day$$

400.0 mg/day exactly. It is below the 584 mg/day Vmax, as every dose rate that reaches a steady state must be.

> **DOES THE NUMBER MAKE SENSE?**
> C_{ss} is about two-thirds of KM + C_{ss} (15 of 21.9), so R is about two-thirds of 584: near 400 mg/day.

Question 2 Worked by the drill

Answer: 6.43 mg/L

WHAT THE STEM GIVES

R1	350 mg	the first daily dose
C_1	9.2 mg/L	steady state on 350 mg
R2	450 mg	the increased daily dose
C_2	20 mg/L	steady state on 450 mg
weight	75 kg	*not needed: the doses stay in mg/day*
age	25 year-old	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$KM = \frac{R2 - R1}{(R1/C_1) - (R2/C_2)}$$, with R1 = 350 mg/day, C_1 = 9.2 mg/L, R2 = 450 mg/day, C_2 = 20 mg/L

Two daily doses in the same patient and the steady-state concentration each produced: the direct method. The 75 kg weight is not needed when the doses stay in mg/day.

ALGEBRA

$$KM = \frac{450 - 350}{(350/9.2) - (450/20)} = \frac{100}{38.04 - 22.50} = \frac{100}{15.54}$$

Each dose over its own concentration: 350 over 9.2 is 38.04 and 450 over 20 is 22.50. Dividing the doses by 75 kg and rounding 4.67 to 4.6 mg/kg/day gives 7.0 mg/L, a rounding error.

ROUND

$$KM = 6.43 \text{ mg/L}$$

(mg/day) over (mg/day per mg/L) leaves mg/L, a concentration, which is the unit of KM. Two decimal places keep the value used for Vmax in question 3.

DOES THE NUMBER MAKE SENSE?

Raising the dose from 350 to 450 mg more than doubled the concentration, from 9.2 to 20 mg/L, the mark of saturation; KM is a concentration: 6.43 mg/L.

Question 3

Worked by the drill

Answer: 595 mg/day

WHAT THE STEM GIVES

KM	6.43 mg/L	from question 2
R2	450 mg	the dose used
C_2	20 mg/L	its own steady state
R1, C_1	350 mg, 9.2 mg/L	the check pair

THE WORKING, ONE LINE AT A TIME

SETUP

$$Vmax = \frac{R(KM + C_{ss})}{C_{ss}}$$

The dose-rate line solved for Vmax: multiply both sides by (KM + C_{ss}), then divide by C_{ss}. Either dose with its own steady-state concentration can be used.

ALGEBRA

$$Vmax = \frac{(450 \text{ mg/day})(6.43 + 20) \text{ mg/L}}{20 \text{ mg/L}} = \frac{11\ 893.5}{20} = 594.7 \text{ mg/day}$$

The second pair, 450 mg/day with 20 mg/L. mg/L cancels and Vmax is in mg/day, the units of the dose rate. A KM rounded too early, such as 7.0 mg/L, gives 607.5 mg/day.

ALGEBRA

Check: $\frac{(350 \text{ mg/day})(6.43 + 9.2) \text{ mg/L}}{9.2 \text{ mg/L}} = 594.6 \text{ mg/day}$

The first pair gives the same value within rounding, because both doses were given to one patient with one Vmax. A disagreement between the two pairs means a dose was paired with the wrong concentration.

ROUND

Vmax = 595 mg/day

594.7 mg/day to the nearest milligram per day is 595 mg/day: the fastest this patient can eliminate the drug, above both the 350 and 450 mg/day doses.

> **DOES THE NUMBER MAKE SENSE?**
> Vmax must exceed 450 mg/day, the higher dose, because that dose reached a steady state: 595 mg/day.

Both dose and concentration pairs give 594.7 mg/day, so 595 mg/day is keyed.

4a

Worked by the drill

Answer: 14.4 mg/L

WHAT THE STEM GIVES

R1	300 mg	the first daily dose
C_1	7.2 mg/L	steady state on 300 mg
R2	500 mg	the increased daily dose
C_2	18 mg/L	steady state on 500 mg
weight	66-kg	*not needed for KM*
V_D	0.7 L/kg	*not needed for KM*

THE WORKING, ONE LINE AT A TIME

SETUP

$KM = \frac{R2 - R1}{(R1/C_1) - (R2/C_2)}$, with R1 = 300 mg/day, C_1 = 7.2 mg/L, R2 = 500 mg/day, C_2 = 18 mg/L

Two daily phenytoin doses in one patient and the steady-state concentration each produced: the direct method. The weight and the volume are not needed for KM.

ALGEBRA

$KM = \frac{500 - 300}{(300/7.2) - (500/18)} = \frac{200}{41.667 - 27.778} = \frac{200}{13.889}$

Each dose over its own concentration, earlier pair first in the denominator. Inverting a ratio, 7.2 over 300, gives a tiny or negative denominator, which signals the slip.

ROUND

KM = 14.4 mg/L

mg/day over (mg/day per mg/L) leaves mg/L, a concentration. KM is the same for both doses because it belongs to the patient and the drug.

DOES THE NUMBER MAKE SENSE?

A two-thirds rise in dose gave a two-and-a-half-fold rise in concentration, so the drug is saturable; KM is a concentration, 14.4 mg/L.

4b

Worked by the drill

Answer: 900 mg/day

WHAT THE STEM GIVES

KM	14.4 mg/L	from part a
R1	300 mg	the dose used
C_1	7.2 mg/L	its own steady state
R2, C_2	500 mg, 18 mg/L	the check pair

THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{Vmax} = \frac{R(KM + C_{ss})}{C_{ss}}$$

The dose-rate line solved for Vmax. With KM known from part a, one dose and the steady-state concentration it produced are enough.

ALGEBRA

$$\text{Vmax} = \frac{(300\ \text{mg/day})(14.4 + 7.2)\ \text{mg/L}}{7.2\ \text{mg/L}} = \frac{6480}{7.2} = 900\ \text{mg/day}$$

The first pair, 300 mg/day with 7.2 mg/L, each from the same dosing period. mg/L cancels and Vmax is in mg/day, the units of the dose rate.

ALGEBRA

$$\text{Check: } \frac{(500\ \text{mg/day})(14.4 + 18)\ \text{mg/L}}{18\ \text{mg/L}} = \frac{16\ 200}{18} = 900\ \text{mg/day}$$

The second pair gives the same 900 mg/day, as it must for one patient with one Vmax. Pairing 300 mg with 18 mg/L would give a different, wrong value.

DOES THE NUMBER MAKE SENSE?

Vmax must exceed the 500 mg/day dose that reached a steady state: 900 mg/day.

4c

Worked by the drill

Answer: 460 mg/day

WHAT THE STEM GIVES

KM	14.4 mg/L	from part a
Vmax	900 mg/day	from part b
C_{ss}	15 mg/L	the target
F	100%	the whole oral dose is available

THE WORKING, ONE LINE AT A TIME

SETUP

$$R = \frac{V_{max} \times C_{ss}}{K_M + C_{ss}}$$

The patient's own KM and Vmax, from parts a and b, with the target steady-state concentration of 15 mg/L in the nonlinear dose-rate line.

ALGEBRA

$$R = \frac{(900 \text{ mg/day})(15 \text{ mg/L})}{(14.4 + 15) \text{ mg/L}} = \frac{13\,500}{29.4} = 459.2 \text{ mg/day}$$

mg/L cancels and leaves mg/day. 15 mg/L lies between the 7.2 and 18 mg/L measured, so the dose lies between 300 and 500 mg/day.

ROUND

$$R = 460 \text{ mg/day}$$

459.2 mg/day rounded to the nearest 10 mg is 460 mg/day, a dose that can be given; the unrounded 459.2 is also accepted.

DOES THE NUMBER MAKE SENSE?

15 mg/L lies between 7.2 and 18 mg/L, so the dose lies between 300 and 500 mg/day: 460 mg/day.

4d

Worked by the drill

Answer: 1.05 days

WHAT THE STEM GIVES

weight	66-kg	$V_D = 0.7 \times 66 = 46.2$ L
V_D	0.7 L/kg	times the weight
C_p	15 mg/L	the concentration the half-life is wanted at
KM	14.4 mg/L	from part a
Vmax	900 mg/day	from part b

THE WORKING, ONE LINE AT A TIME

UNIT

$$V_D = 0.7 \text{ L/kg} \times 66 \text{ kg} = 46.2 \text{ L}$$

The half-life line needs the volume in litres, and the stem gives it per kilogram, so it is multiplied by the 66 kg weight first.

SETUP

$$t_{½} = V_D\left(\frac{0.693}{V_{max}}\right)(K_M + C_p)$$

For a saturable drug the half-life depends on the concentration, so it is calculated at the stated C_p of 15 mg/L, not from 0.693 over a fixed k.

ALGEBRA

$$t_{½} = 46.2 \text{ L} \times \frac{0.693}{900 \text{ mg/day}} \times (14.4 + 15) \text{ mg/L} = 46.2 \times 0.00077 \times 29.4 = 1.046 \text{ days}$$

L × mg/L leaves mg on top, which cancels the mg of mg/day and leaves days. Leaving V_D in L/kg gives 0.016, a result per kilogram, not a time.

ROUND

$t½ = 1.05$ days

1.046 days to two decimal places. At 7.2 or 18 mg/L the half-life comes out different, because KM + C_p changes with the concentration.

> **DOES THE NUMBER MAKE SENSE?**
> Vmax is in mg/day, so the half-life comes out in days: 1.046 days, about one day, not a fraction of an hour.

4d

Worked by the drill

Answer: 30.6 L/day

WHAT THE STEM GIVES

C_p	15 mg/L	added to KM in the denominator
KM	14.4 mg/L	from part a
Vmax	900 mg/day	from part b
V_D	0.7 L/kg	*not needed for clearance*

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_T = \frac{Vmax}{KM + C_p}$$

For a saturable drug clearance depends on the concentration, which sits in the denominator, so it is calculated at the stated 15 mg/L.

ALGEBRA

$$Cl_T = \frac{900 \text{ mg/day}}{(14.4 + 15) \text{ mg/L}} = \frac{900}{29.4} = 30.61 \text{ L/day}$$

mg/day over mg/L leaves L/day, a volume per unit time, which is what a clearance is. Converting to L/hr is allowed, but L/day is accepted as it stands.

ROUND

$Cl_T = 30.6$ L/day

30.61 L/day to one decimal place. The same volume and clearance give the half-life of part d: 0.693 × 46.2 L over 30.6 L/day is 1.05 days.

> **DOES THE NUMBER MAKE SENSE?**
> At 15 mg/L, KM + C_p is 29.4 mg/L, about twice KM, so clearance is well below Vmax ÷ KM: 30.6 L/day.

4e

Worked by the drill

Answer: 11.5 mg/L

WHAT THE STEM GIVES

R	400 mg daily	the new dose rate
KM	14.4 mg/L	from part a

Vmax	900 mg/day	from part b

THE WORKING, ONE LINE AT A TIME

SETUP

$$R \times KM + R \times C_{ss} = Vmax \times C_{ss}, \text{ so } C_{ss} = \frac{R \times KM}{Vmax - R}$$

The dose-rate line rearranged for C_{ss}: multiply out the denominator, gather the C_{ss} terms on one side, factor out C_{ss}, and divide by (Vmax − R).

ALGEBRA

$$C_{ss} = \frac{(400 \text{ mg/day})(14.4 \text{ mg/L})}{(900 - 400) \text{ mg/day}} = \frac{5760}{500} = 11.52 \text{ mg/L}$$

mg/day cancels and leaves mg/L. The denominator is Vmax minus the dose, 500 mg/day; adding them instead gives 4.4 mg/L, which is below the 7.2 mg/L the 300 mg dose produced.

ROUND

$$C_{ss} = 11.5 \text{ mg/L}$$

11.52 mg/L to one decimal place. 400 mg/day lies between the 300 and 500 mg/day doses, and 11.5 mg/L lies between their 7.2 and 18 mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> 400 mg/day lies between 300 and 500 mg/day, so C_{ss} lies between 7.2 and 18 mg/L: 11.5 mg/L.